Ring-LWE and Ideal Lattice Cryptography

Algebraic Number Theory Meets Post-Quantum Security

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Motivation

Classical public-key cryptography (RSA, ECC) will be broken by Shor's algorithm on a quantum computer. **Lattice-based cryptography** offers post-quantum security whose hardness reduces to worst-case geometric problems on lattices—problems believed hard even for quantum adversaries. Ring-LWE combines this hardness guarantee with the algebraic structure of number rings to achieve practical efficiency.

Step 1 — Lattices and Geometry of Numbers

A lattice $\mathcal{L} = \bigoplus_i \mathbb{Z}b_i \subset \mathbb{R}^n$ is a discrete subgroup. The central hard problem is $\mathbf{SVP}_\gamma$: find a shortest nonzero vector. Minkowski's theorem gives $\lambda_1(\mathcal{L}) \le \sqrt{n} \cdot \det(\mathcal{L})^{1/n}$, connecting lattice geometry to the course framework.

Step 2 — Number Rings as Ideal Lattices

Let $K = \mathbb{Q}(\zeta_n)$, $n = 2^k$, with ring of integers $\mathcal{O}_K = \mathbb{Z}[\zeta_n]$ (Dedekind domain). The **canonical embedding** $\sigma : K \hookrightarrow \mathbb{R}^n$ maps any ideal $\mathfrak{a} \trianglelefteq \mathcal{O}_K$ to a full-rank lattice—an **ideal lattice**—with

$$\det(\sigma(\mathfrak{a})) = \mathcal{N}(\mathfrak{a}) \cdot \sqrt{|\Delta_K|}, \qquad \Delta_K = \mathcal{N}(\mathfrak{d}_{K/\mathbb{Q}}).$$

This connects to course Objectives 1–4: number rings, Dedekind ideals, prime splitting, norms, and discriminants.

Step 3 — LWE and Its Ring Variant

LWE (Regev 2005): given $(a_i,\ \langle a_i, s\rangle + e_i) \in \mathbb{Z}_q^n \times \mathbb{Z}_q$ with small errors e_i, recover s. Solving LWE on average is as hard as SVP in the worst case (quantum reduction). Cost: $O(n^2)$ key size.

Ring-LWE (Lyubashevsky–Peikert–Regev 2010): lift LWE to $R_q = \mathbb{Z}[x]/(\Phi_n(x), q)$. Samples become

$$(a,\ a \cdot s + e) \in R_q \times R_q^\vee, \qquad R^\vee = \mathfrak{d}_{K/\mathbb{Q}}^{-1},$$

where $R^\vee$ is the inverse different (course Obj. 4). Ring multiplication is negacyclic convolution, computed via NTT in $O(n \log n)$. The prime splitting of q in $\mathcal{O}_K$ (course Obj. 3) determines the CRT structure of R_q.

Hardness: decision Ring-LWE $\le_{\text{poly}}^{\text{quantum}}$ Ideal-SVP$_{\tilde{O}(n/\alpha)}$.

Step 4 — Applications and Open Problems

NIST Standards (2024)

- FIPS 203 (ML-KEM / Kyber): key encapsulation
- FIPS 204 (ML-DSA / Dilithium): digital signature
- Keys <2 KB; security $\approx$ AES-128

Open Problems

- Quantum hardness of Ideal-SVP vs. SVP
- Subfield attacks via Galois group of $K/\mathbb{Q}$
- Security of non-cyclotomic rings

Course Connections. §2: number rings & discriminants (Obj. 1,4); §2: Dedekind ideals & norms (Obj. 2); §3: prime splitting in R_q (Obj. 3); §2: Minkowski / geometry of numbers (Obj. 5).

Key References. O. Regev, *J. ACM* **56**(6), 2009. Lyubashevsky–Peikert–Regev, *J. ACM* **60**(6), 2013. C. Peikert, *Found. Trends TCS* **10**(4), 2016. NIST FIPS 203/204, 2024.